



GE Medical Systems

Technical Publication

Direction 2340902-100

Revision 0

Service Introduction Planner (iSERV)

HiSpeed QX/i

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REVISION HISTORY

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Revision	Date	Reason for change
0	06/20/02	Initial Release

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SERVICE INTRODUCTION PLANNER

The information contained in this document has been provided to assist the GE Service Engineer and Service Operation group with the introduction of the new Global Console with Octane II.

This document is Not intended to replace service manuals or user guides. It is a collection of information taken from several documents. ALL of which are available and should be used to Install, Operate, and Service the HiSpeed QX/i system.

1. INTRODUCTION

This document provides information on HiSpeed QX/i to be phased into the CT Market Q2, 2002.

The HiSpeed QX/i is the 4slice CT scanner system which defeatures some of the specifications from the LightSpeed Plus MDAS system for offering lower price range.

The HiSpeed QX/i also features the new global console. The Gantry, Table and PDU are almost same as the current the LightSpeed Plus/Ultra MDAS system.

M3 and M4 are scheduled in FW22 and FW38 respectively.

For the details of the new global console, refer to 2344406-100, Global Console(Octane II) Introduction Planner.

2. SPECIFICATIONS

Cat#	B76842AB	B76822AB	B76832AB
Market	Japan	EXP (GEMSA,GEMSAm)	EXP (GEMSE)
# of slices	4slice	4slice	4slice
Scan speed	0.7,1.0,2.0,3.0,4.0sec	0.7,1.0,2.0,3.0,4.0sec	0.7,1.0,2.0,3.0,4.0sec
Tube	6.3MHU	6.3MHU	6.3MHU
Max. mA	440mA	350mA	350mA
Max. Power	53.2KW	42KW	42KW
kVA	90kVA	90kVA	90kVA
Cooling algorithm	6.3MHU	6.3MHU	6.3MHU
Helical memory	120sec	120sec	120sec
Recon time	1.5sec	1.5sec	1.5sec
Max. Scanable range	160cm	160cm	160cm
Gantry controllers	Front	Front	Front
In Room Start	Std.	Std.	Std.
Helical Tilt	Std.	Std.	Std.
Keyboard	Std.	Std.	*Opt.

*European language keyboard is optional.

3. FEATURES

The followings are major changes and de-features from LightSpeed Plus/Ultra MDAS system.

- New console(Global Console with Octane II) and software(LS2002)
- The color of the Gantry Top, side, bottom covers, Table bottom covers, Cradle handle and Cradle mats will be changed to gray.
- Scan Time: 0.7sec as standard
- Max mA: 350mA as standard
- Recon Time: 1.5sec as standard
- Scannable Range: 150cm in Helical/160cm Axial (This is made by short Cradle Extender.)
- Rear side controller is optional

4. SOFTWARE

Below is a matrix of the part numbers of the software for HiSpeed QX/i sytems (at M3).

HiSpeed QXI Serialized Set w Rtg PI	2344273	Shipped with system
HiSpeed QXI Serialized OS and Apps	2344272	Shipped with system
Pacific Star Application SW (Ver. 100.3_HSQXI_M3)	2343153	Shipped with system
Irix64 ver6.5.11 Disk1 ver2	2305087-2	Shipped with system
H3 Irix64 ver6.5.11 Disk2	2305088	Shipped with system
H3 Irix64 ver6.5.11 Disk3	2305089	Shipped with system
Patch Software	2345869	Shipped with system
Rating Plate	46-302200P2	Shipped with system
Service Software (Class C)	2344422	Distributed by Tab 182
iLINQ Software (Class M)	2344423	Distributed by Tab 182

Note: Patch Software CDROM(2345869) will be marged into one Application Software CDROM sometime after M3.

The part numbers of the Application CDROM and OS & Application CDROM set will be changed as follows,

- Pacific Star Application SW (Ver.100.3_HSQXI_M3.1) : 2348417
- HiSpeed QXI Serialized OS and Apps : 2348418

5. PRE-INSTALLATION

Pre-installation information can be found in the Pre-Installation manual ref. 2340894-100.

6. INSTALLATION

6.1 INSTALLATION TIME

Same as LightSpeed Plus. The system can be installed in 30 man-hours (without Options and Travel Time). Mechanical installation requires two people and electrical one.

6.2 DOCUMENTS

Pre-Installation Manual (2340894-100), distributed through TAB (#88)

Installation Manual (2340895-100), paper manuals shipped with the system

6.3 TOOLS

WRENCHES:

Standard combination wrench set

Metric combination wrench set

Metric Hex Key (Allen wrench) set

Standard Hex Key (Allen wrench) set

Calibrated 1/2 " and 3/8" drive torque wrench:

0-100 N-m (0-100 ft.-lb.)

SOCKETS AND EXTENSIONS:

1/2" drive ratchet wrench

3/8" drive ratchet wrench

1/2" drive 3" long extension

1/2" drive 6" long extension

3/8" drive 12" long extension

Standard 3/8 " drive socket set

Metric 3/8" drive socket set

3/4" deep well socket 3/8 "

1-1/8" socket 1/2" drive

1-1/2" socket 1/2" drive

3/8 " drive universal joint

1/4" or 3/8" drive hex bit set

3/8" to 1/2 " socket adapter

3" X 3/8" drive drill adapter

Metric hex bit set 1/4" Or 3/8" drive

10 & 14 MM Hex bit for 3/8' drive

SCREWDRIVERS:

Phillips screwdrivers (small, medium, and large)

Straight blade screwdrivers (small, medium, and large)

Drill/driver hex bit set

DRILL BITS:

Set of standard drill bits

1/2" masonry bit, minimum, 8" in length

13mm masonry bit, min. 8" in length

POWER TOOLS:

3/8" & 1/2" drill, cordless or electric

Reciprocating Saw (Sawzall or equivalent)

*Hammer Drill with 3 or 4 point 1/2" X 8" bit

Assorted blade set for Reciprocating saw

*Sears™ 17740 Shop vacuums with (2) 25' extension cords

MISCELLANEOUS

Vacuum Filters - HEPA or dry wall dust filter (Sears 17918) or equivalent

• HEPA stands for High Efficiency Particle Arresting

7. PLANNED MAINTENANCE

Planned Maintenance schedules for the system contained on Revision 14 or greater of the PM Manual (2141834-100). Maintenance Frequency depends on Site activity.

Recommended Frequency is:

- 4 PM visits/year (Systems In-warranty)
- 1PM per 250 K slices (systems Out of warranty)

Required Time: 13 hours/year (In warranty)

8. CORRECTIVE MAINTENANCE

MTBF (Estimate) : 33 days at M4 (Same as LightSpeed Plus (MDAS))

M-WTTR (Estimate) : 5 hours (Same as LightSpeed Plus (MDAS))

Tube Change : Same as LightSpeed Plus (MDAS)

Detector Change : Same as LightSpeed Plus (MDAS)

9. SERVICE TOOLS

9.1 ALIGNMENTS

5mm hex bit for 3 / 8 " drive 6"

2.5mm Allen hex bit for 1 / 4 "

long (Snap-on™ FAML5E or equivalent)

Standard FE Tool Kit

10mm hex bit for ratchet wrench (ball end)

14 & 10mm ball hex socket bit for 3/8" drive

(Snap-on FABM10E or equivalent)

14 & 10mm hex socket bit for 1/2" drive

10mm open-end thin wrench (Snap-on SRSM10 or equivalent)

21mm open-end thin wrench (Snap-on LTAM2124 or equivalent)

Electrical Tools: (Must be calibrated yearly)

Fluke 87 DVM or equivalent

Clamp on amp meter

Ground Rod Tester Clamp on Amp Probe

GE HV Bleeder

Scope with 10X probes

Beckman CT231 clamp-on Amp probe: 46-194427P228 (Fits up to #2 size wire)

Beckman CT232 clamp-on Amp probe: 46-194427P270 (Fits #1 size wire and larger)

ECOS 1023H Power Line-Leakage Current Tester (60HZ): 46-194427P248

Adapter Cables for Power Line-Leakage Current Tester: 46-194427P249

9.2 ALIGNMENT & CALIBRATION TOOLS:

Polaroid Film and Holder for type 52 film

Caliper Dial indicators

Alignment dial Gauge set

Torque Wrench kit

Dial indicator mount

Polaroid type 52 film

Scope Probe-to-bleeder Cable 46-219921G1

Spanner Wrench 46-404295g1

LOTO kit

9.3 DETECTOR SERVICE TOOLS:

2.5 mm hex bit

Canned Air (Grounded) with metal nozzle (Miller
Stephenson Aero Duster MS-222N or equivalent)

Ground Wrist Strap

Dental Pick

Anti-Static Bags

Grounded Mat

Electric Screw Driver

Anti-static finger cots (part number 2221235 [small] and
2221235-2 [large])

10. PRODUCT CATALOGS

*) This Cat# may be changed near future due to transition to Oracle system.

			2342364	2342365	2342366
			B76842AB	B76822AB	B76832AB
Name	Cat#	GPN#	Japan	GEMSA GEMSAm	GEMSE
Sub System & Collectors					
Gantry	B78122AA	2342153	Std.	Std.	Std.
Table	B7850PT	2276776-3	Std.	Std.	Std.
OC (GOC1-Octane II)	B7858L	2341104	Std.	Std.	Std.
PDU	B7850NP	2269902	Std.	Std.	Std.
System SW and Manual Collector	B78212AB	2342677	Std.	Std.	Std.
System Shipping Collector, Gray	B78222AB	2281302-2	Std.	Std.	Std.
System Long Cable Collector	B7850LL	2281840	Std.	Std.	Std.
Hardware Options					
21" CRT monitor x 2	B7850MM*	2333243	Opt.	Opt.	Opt.
18" LCD monitor x 2	B7858LC*	2309592	Opt.	Opt.	Opt.
DASM (Analog)	B7720W	2276650	Opt.	Opt.	Opt.
DASM (Digital)	B7700L	2213357	Opt.	Opt.	Opt.
Rear gantry controllers	B76932AB	2344575	Opt.	Opt.	Opt.
Axial Head Holder (Deep)	TBD		Opt.(at M4)	Std.	Std.
Axial Head Holder (Shallow)	TBD		Std.	Opt.(at M4)	Opt.(at M4)
English Keyboard&SCIM	B7800KE*	2275758	Std.	Std.	Opt.
French Keyboard&SCIM	B7800KF*		-	-	Opt.
German Keyboard&SCIM	B7800KG*		-	-	Opt.
Scandinavian Keyboard&SCIM	B7800KS*		-	-	Opt.
Software Options					
Helical Tilt	B7810TL	2305990	Std.	Std.	Std.
0.7sec Scan	B7868SP	2354410	Std.	Std.	Std.
Thin Twin Helical	B7810TH	2305988	Std.	Std.	Opt.
Smart Prep	B7530SP		Std.	Std.	Opt.
53.2KW(440mA) Power option	B7868PW	2345411	Std.	Opt.	Opt.
0.5sec. Recon option	B7810RP	2305991	Opt.	Opt.	Opt.
ConnectPro	B7500PN		Opt.	Opt.	Opt.
Bar Code	B7540RB		Opt.	Opt.	Opt.
3D analysis	B7710MD		Opt.	Opt.	Opt.
Navigator	B7540N	2149345	Opt.	Opt.	Opt.
DentaScan	B7540SD	2149346	Opt.	Opt.	Opt.

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SmartScore Pro	B7850KC		Opt.	Opt.	Opt.
CT Perfusion2	B7700PS	2277298-2	Opt.	Opt.	Opt.
SmartStep w/o boom	TBD		Opt.(at M4)	Opt.(at M4)	Opt.(at M4)
SmartStep w/ boom	TBD		Opt.(at M4)	Opt.(at M4)	Opt.(at M4)
Other Options					
Remote 20 inch Monitor	B7530RC	2158191	Opt.	Opt.	Opt.
Dolley	B78502LD	2342349	-	Opt.	Opt.
SmartStep w/o boom	TBD		Opt.(at M4)	Opt.(at M4)	Opt.(at M4)
SmartStep w/ boom	TBD		Opt.(at M4)	Opt.(at M4)	Opt.(at M4)

11. SERVICE AND OPEARTOR DOCUMENTATION

Document Title	Direction No.	Class
For System		
General Class Service Documentation Collector	2342051	A
Class A CD-ROM	2340891-200	A
Class M CD-ROM	2340892-200	M
Pre-Installation	2340894-100	A
Installation	2340895-100	A
Software Installation Procedures	2340896-100	A
Service Manual - General	2340897-100	A
Service Manual – Adv.	2340898-100	C
Illustrated Parts List	2340899-100	A
Schematics, Boards and Diagrams	2340900-100	A
Conformance Statement	TBD	A
CT PM Procedures	2141834-100	A
LightSpeed Family Detector Change Procedure	2335850-100	A
MT5634 (2245794) Multi-Tech Modem - User Manual	2262950-100	A
HiSpeed QX/i Service Introduction Planner	2340902-100	M
CT iLinq Manual	2243323-100	M
CT Software Cross Reference Class C, M, & Application	2328773-100	M
For Options		
Options Installation	2211224-100	A
For Operators		
Technical Reference	2344293-100	A
Learning and Reference Guide	23418-9-200	A

Class A: Documentation shipped with every system

Class C: Licenced documentation

Class M: Proprietary (GEMS only) documentation

12. SERVICE DOCUMENTATION DISTRIBUTION SYSTEMS

12.1 INTRANET

GEYMS Technical Publications Web Site: <http://3.36.10.5/>

12.2 CD-ROM

Class M CD-ROM2340892-200, Revision 0(to be released FW26/2002) will contain all the new CT HiSpeed QX/i Service manuals listed above.

12.3 TAB

Tab 182 will be used for distributing HiSpeed QX/i Service Manuals and Service Software.

13. TRAINING

Pilot Training

Course Number - CT009

The training for the introduction of the HiSpeed QX/i will be done as a part of the Global Console I training.

This training is planned to be in the form of a Power Point presentation that will be available via the TEDS website. There will be several T-Con sessions coordinated to support the discussion of this training information.

Information about these T-Cons will be shared via the weekly QMI call and will be posted to the TEDS website.

Post M4 Training

eLearning Module(CDROM) will be provided as a continuing education product.

Note: As a continuing education product, all students are required to demonstrate a familiarity with the LightSpeed Ultra configuration of products to take advantage of the new offering.

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